

Conium maculatum (Apiaceae)

Hemlock

Description Graceful biennial growing to a height of 8 ft (2.5 m). Has slender, red-speckled stems, finely divided leaves, small clusters of white flowers, and small seeds that have beaded ridges.



Hemlock is highly poisonous and was used for capital punishment.

Habitat & Cultivation Commonly found in Europe, hemlock also grows in temperate regions of Asia and North America. It flourishes in damp meadows, on riverbanks, and in open areas. The seeds are gathered when almost ripe in summer.

Parts Used Leaves, seeds.

Constituents Hemlock contains alkaloids, mainly coniine, and a volatile oil. Coniine is extremely toxic and causes congenital deformities.

History & Folklore Hemlock is notorious as the poison administered as a capital punishment in ancient Greece. The Greek philosopher Socrates died in 399 BCE after drinking hemlock juice. According to an old English tradition, the stems took their color in sympathy with the mark placed on Cain's forehead after he murdered Abel. In the 19th century, hemlock was used in conventional medicine as a painkiller.

Medicinal Actions & Uses In extremely small quantities, hemlock is sedative and analgesic; in larger doses it causes paralysis and death. Rarely used today, it has been prescribed in the past as a treatment for epilepsy, Parkinson's disease, and Sydenham's chorea. Hemlock has also been used to treat acute cystitis.

⚠ Cautions Do not take internally. Use externally only under professional supervision. Hemlock is subject to legal restrictions in many countries.

Convallaria majalis (Liliaceae)

Lily of the Valley

Description Attractive perennial growing to 9 in (23 cm). Has a pair of elliptical leaves, clusters of bell-shaped white flowers on one side of the stem, and red berries.

Habitat & Cultivation Native to Europe, this herb is also distributed over North America and northern Asia. It is widely cultivated as a garden plant. The leaves and flowers are gathered in late spring as the plant comes into flower.

Parts Used Leaves, flowers.

Constituents Lily of the valley contains cardiac glycosides, including the cardenolides convallotoxin, convallaside, convallatoxol, and others, and flavonoid glycosides. The cardiac glycosides act to strengthen a weakened heart.

History & Folklore The herbalist Apuleius, writing in the 2nd century CE, records that Apollo gave lily of the valley as a gift to Asclepius, the god of healing. In the 16th century, the herbalist John Gerard had the following to say about its therapeutic value: "The flowers of the valley lillie distilled with wine, and drunke to the quantitie of a spoonful, restore speech unto those that have the dumb palsie and that are fallen into apoplexy, and are good against the gout, and comfort the heart."

Medicinal Actions & Uses Lily of the valley is used by European herbalists in place of common foxglove (*Digitalis purpurea*, p. 202). Both herbs have a profound effect



Lily of the valley encourages a regular heartbeat and acts as a strong diuretic.

in cases of heart failure, whether due in the long term to a cardiovascular problem, or to a chronic lung problem such as emphysema. Lily of the valley encourages a failing heart to beat more slowly and regularly, and to pump more efficiently, thereby improving blood flow to the heart itself via the coronary arteries. It is also diuretic and lowers blood volume. The herb is better tolerated than foxglove, as it does not accumulate within the body to the same degree. Relatively low doses are required to support heart rate and rhythm, and to increase urine production.

⚠ Cautions Use only under professional supervision. Lily of the valley is subject to legal restrictions in some countries.

Conyza canadensis syn. *Erigeron canadensis* (Asteraceae)

Canadian Fleabane

Description Erect annual herb growing to 3 ft (1 m). Has narrow, dark green, lance-shaped leaves and clusters of small white flower heads that quickly fade into silky white tufts.

Habitat & Cultivation Native to North America, Canadian fleabane is now common in South America and Europe. It thrives on uncultivated and recently cleared land, often invading in large swaths. It is gathered from the wild when in flower.

Parts Used Aerial parts.

Constituents Canadian fleabane contains a volatile oil (including limonene, terpineol, and linalool), flavonoids, terpenes, plant acids, and tannins.



Canadian fleabane was used both ritually and medicinally by Native North Americans.